

THE FIVE PHYSICAL COLUMNS OF EMBODIED INTELLIGENCE

The Fundamental Physical Bounds Governing Every Closed-Loop Cyber-Physical System

COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 5
TIME & FRESHNESS τ_{world} Latency Bounds • Sense-to-actuation delay Δt • P99.9 tail latency freshness • Stale observations cause drift • Hard deadlines: $\Delta t \leq \tau_{\text{world}}$	INERTIA & MOMENTUM $p = mv$ Kinetic Energy • Mass cannot stop instantaneously • Kinetic braking distance d_{stop} • Centrifugal & Coriolis forces • Work envelope geofencing	ACTUATION DYNAMICS τ_{motor} & Friction Limits • BLDC torque-speed envelopes • Gearbox backlash & jerk bounds • Coulomb/viscous joint friction • Smooth C^2 acceleration profiles	ENERGY & THERMODYNAMICS $P = IV$ & Joule Heat • Battery voltage sag under load • Junction heating $T_j = T_a + P \cdot \theta_{JA}$ • DVFS thermal throttling cascades • Regenerative braking limits	SILICON & COMPUTE SRAM, AXI & DMA Memory • Zero dynamic malloc in RT loops • UMA crossbar bus contention • Hardware watchdog supervisors • Deterministic FreeRTOS priority

FIRST-PRINCIPLES PHYSICAL GUARANTEE

Software models propose intent; the Five Physical Columns enforce the boundary of physical possibility.